

Complex Analysis Exam Problems

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Fall 2025

Part I

1. Define $\gamma: [0, 2\pi] \rightarrow \mathbb{C}$ by

$$\gamma(t) = e^{it}.$$

Compute

$$\int_{\gamma} z^n dz$$

for every integer n .

2. Let γ be the closed polygon

$$[1 - i, 1 + i, -1 + i, -1 - i, 1 - i].$$

Find

$$\int_{\gamma} \frac{1}{z} dz.$$

3. 1. Show that a Möbius transformation has 0 and ∞ as its only fixed points if and only if it is a dilation, but not the identity.
2. Show that a Möbius transformation has ∞ as its only fixed point if and only if it is a translation, but not the identity.
4. Let G be a region, and suppose that $f: G \rightarrow \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

5. Let

$$f(z) = e^x \sin y + ie^x \cos y, \quad z = x + iy.$$

Determine whether f is analytic on the complex plane.

6. Map

$$G = \mathbb{C} \setminus \{z : -9 \leq z \leq 9\}$$

onto the open unit disk by an analytic function f . Can f be one-to-one?

7. Give the principal branch of

$$\sqrt{1 - z^3}.$$

Part II

8. Compute the integral

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz,$$

where γ is the closed path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi.$$

9. Compute the integral in the previous exercise with γ the closed path

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$